Yicheng Gu

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Key Skills

Solid Domain Knowledge: Extensive knowledge of Deep Learning and Digital Signal Processing; specifically applied to Neural Vocoder, Neural Audio Codec, and Digital Audio Effects.
Multi-disciplinary: Experience encompassing Research, Open Source Systems, Dataset Development, DJ, Commercial Sound Design, and Music Production.

Education

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
Bachelor of Engineering in Computer Science and Engineering — *Sep. 2022 – Present*

- GPA: 3.965 / 4.0, Ranking: 1 / 320

Aalto University — Espoo, Finland
Exchange Student in Computer Science — *Sep. 2024 – Jun. 2025*

- GPA: 4.75 / 5.0

Experience

Sizigi Studio — Tokyo, Japan
Audio Engineer, Spellbrush — *Jun. 2025 – Present*
Project Leader: Tiger Tang

- Text-To-Speech Synthesis
  - Built the TTS system for Nori, the world's first AI Waifu.
- High-Fidelity Neural Audio Synthesis
  - Built the first aliasing-free neural audio vocoder and codec systems (Pupu-Vocoder and Pupu-Codec) in the whole research area .
- Self-Supervised Music Representation Learning
  - Adapted the recent progress in the area of vision self-supervised representation learning (SSL) into music, built the first contrastive-learning-based music SSL model in the time-frequency domain, and achieved the state-of-the-art (SOTA) performance.

Human Language Technology Lab — Shenzhen, China
Research Assistant, School of Data Science, CUHK-Shenzhen — *Oct. 2022 – Present*
Supervisor: Prof. Zhizheng Wu

- Singing Voice Conversion
  - Investigated characteristics and the complementary role of different Content-Based Features for the Singing Voice Conversion system.
- Speech Synthesis
  - Proposed a large-scale and diversified dataset, Emilia, to facilitate speech research, which has been utilized by various international institutions.
- Neural Vocoder
  - Built a Discriminator based on the Constant-Q Transform (CQT) and Continuous Wavelet Transform (CWT) to improve the Vocoder's synthesis quality. The methods have been implemented and supported by NVIDIA BigVGAN .

Acoustic Lab Espoo, Finland
Visiting Scholar, DICE, Aalto University *Sep. 2024 – Jun. 2025*
Supervisor: Prof.Lauri Juvela

- Digital Audio Effects
  - Proposed the SOTA Neural Autotune benchmarking against Melodyne.
  - Proposed the first large-scale and diverse dataset for Virtual Analog Modeling.

Shanghai AI Laboratory Shanghai, China
Research Assistant *Dec. 2023 – Mar. 2025*
Supervisor: Dr. Yanhong Zeng

- Video-to-Audio Generation
  - Integrated IP-Adapter and Sound Event Detection model to existing Audio Generation pipeline, obtaining both Audio-Visual Synchronization and Text-Controllability .

Moises Utah, US
Research Collaboration *Sep. 2025 – Present*
Collaborator: Joaquim Cavalcante

- Neural Autotune
  - Built an aliasing-free neural pitch manipulation system and an autotune system that can automatically correcting the pitch contour of the singing voice based on the backing tracks.

Projects

Amphion
An Open-Source Audio, Music and Speech Generation Toolkit

- Migrate and adapt various well-known, widely used, or SOTA vocoders into our system.
- Integrate comprehensive Objective Evaluation Metrics into the framework.
- Conducted comprehensive TTS experiments for the Emilia dataset.

Publications

Yicheng Gu, Junan Zhang, Chaoren Wang, Jerry Li, Zhizheng Wu, Lauri Juvela, ”Aliasing-Free Neural Audio Synthesis,” Submitted to TASLP.

Yicheng Gu, Xueyao Zhang, Liumeng Xue, Haizhou Li, Zhizheng Wu, ”An Investigation of Time-Frequency Representation Discriminators for High-Fidelity Vocoder,” TASLP.

Yicheng Gu, Xueyao Zhang, Liumeng Xue, Zhizheng Wu, ”Multi-Scale Sub-Band Constant-Q Transform Discriminator for High-Fidelity Vocoder,” ICASSP 2024.

Xueyao Zhang*, Liumeng Xue*, **Yicheng Gu***, Yuancheng Wang*, et al., ”Amphion: An Open-Source Audio, Music and Speech Generation Toolkit,” SLT 2024.

Haorui He*, Zengqiang Shang*, Chaoren Wang*, Xuyuan Li*, **Yicheng Gu**, et al., “Emilia: An Extensive Multilingual and Diverse Speech Dataset for Large-Scale Speech Generation,” SLT 2024.

Haorui He*, Zengqiang Shang*, Chaoren Wang*, Xuyuan Li*, **Yicheng Gu**, et al., “Emilia: A Large-Scale, Extensive, Multilingual, and Diverse Dataset for Speech Generation,” TASLP.

Yiming Zhang, **Yicheng Gu**, Yanhong Zeng, et al., ”FoleyCrafter: Bring Silent Videos to Life with Lifelike and Synchronized Sounds,” IJCV.

Yicheng Gu, Chaoren Wang, Zhizheng Wu, Lauri Juvela, ”Neurodyne: Neural Pitch Manipulation with Representation Learning and Cycle-Consistency GAN,” Interspeech 2025.

Yicheng Gu*, Runsong Zhang*, Lauri Juvela, Zhizheng Wu, ”Diff-SSL-G-Comp: Towards a Large-Scale and Diverse Dataset for Virtual Analog Modeling,” DAFx 2025.

Xueyao Zhang, **Yicheng Gu**, et al., ”Leveraging Content-based Features from Multiple Acoustic Models for Singing Voice Conversion,” Machine Learning for Audio Workshop at NeurIPS 2023.

Xueyao Zhang, Zihao Fang, **Yicheng Gu**, et al., ”Leveraging Diverse Semantic-based Audio Pretrained Models for Singing Voice Conversion,” SLT 2024.

Yicheng Gu*, Chaoren Wang*, Junan Zhang*, Xueyao Zhang, et al., ”SingNet: Towards a Large-Scale, Diverse, and In-the-Wild Singing Voice Dataset,” Submitted.

Zirui Li, Jens Edlund, **Yicheng Gu**, Nhan Phan, Lauri Juvela, Mikko Kurimo, ”Nord-Parl-TTS: Finnish and Swedish TTS Dataset from Parliament Speech,” Submitted to ICASSP 2026.

Honors and Awards

- The Academic Performance Scholarship, Class A (Top 1%, 2025)
- The Nobel Class (Top 1, 2024)
- The Academic Performance Scholarship, Class A (Top 1%, 2024)
- The Academic Performance Scholarship, Class B (Top 3%, 2023)
- “LanHuaYing” Scholarship (Top 10 admitted students in Zhejiang Province, 2022)
- University Entrance Scholarship (Top 1% in Zhejiang Province, 2022)

Cross-disciplinary Abilities

Music Production: experienced in Composing, Mixing, and Mastering for different genres including Pop, Electronic, etc.
Sound Design: experienced in Sound Designing for movie, game, etc.